

Emotion Recognition from Speech

CNN vs. LSTM: A Comparative and Explainability Study

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The project

SER (Speech Emotion Recognition) on RAVDESS dataset

Model Comparison

Compare a CNN-based and an LSTM-based model for the same task

Dataset Strategy

Speaker Independent dataset split

Explainability

Focus on understanding model decisions and underlying patterns

Evaluation Metrics

- Confusion matrix & Accuracy
- Macro F1 score
- Per class F1 score

State Of The Art

[1] Hand crafted features + DNN (81.2%)

→ no speaker-ind. split

[2] Wav2Vec with dense layer + LSTM model (74.6%)

→ speaker ind. split

[3] CNN (75%) + Attention Mechanism (85%)

→ no speaker-ind. split

[4] 3D CNN + LSTM model (96.18%)

→ no speaker-ind. split

[5] CNN + Attention Mechanism + Augmentation (81%)

→ no speaker-ind. split

The Data

RAVDESS Dataset

- **1,440** .wav files
- **24** Actors (12 male, 12 female)
- **8** Emotional states

Filename Metadata Structure

03-01-06-01-02-01-12.wav

- Emotion
- Statement
- Actor* → 60 recordings each

Emotion Encoding

01: Neutral	x96	recordings
02: Calm	x192	
03: Happy	x192	
04: Sad	x192	
05: Angry	x192	
06: Fearful	x192	
07: Disgust	x192	
08: Surprised	x192	

*Actors: Odd = Male, Even = Female



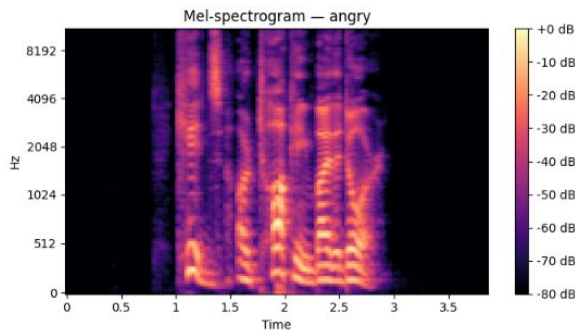
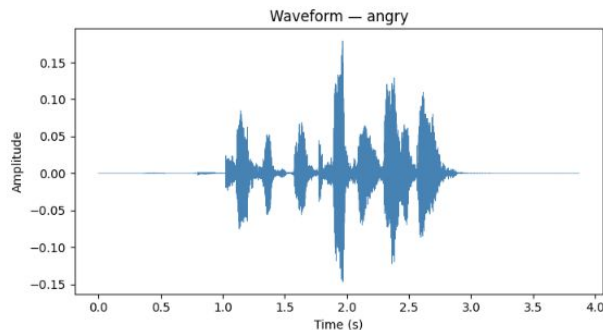
Speaker Independent Split

The Data

CNN Pipeline: Mel Spectrogram

Audio (.wav) → pad/truncate (3s) → waveform → 2D Spectrogram + Mel Scale

Output: Mel Spectrogram of shape (128, 130)

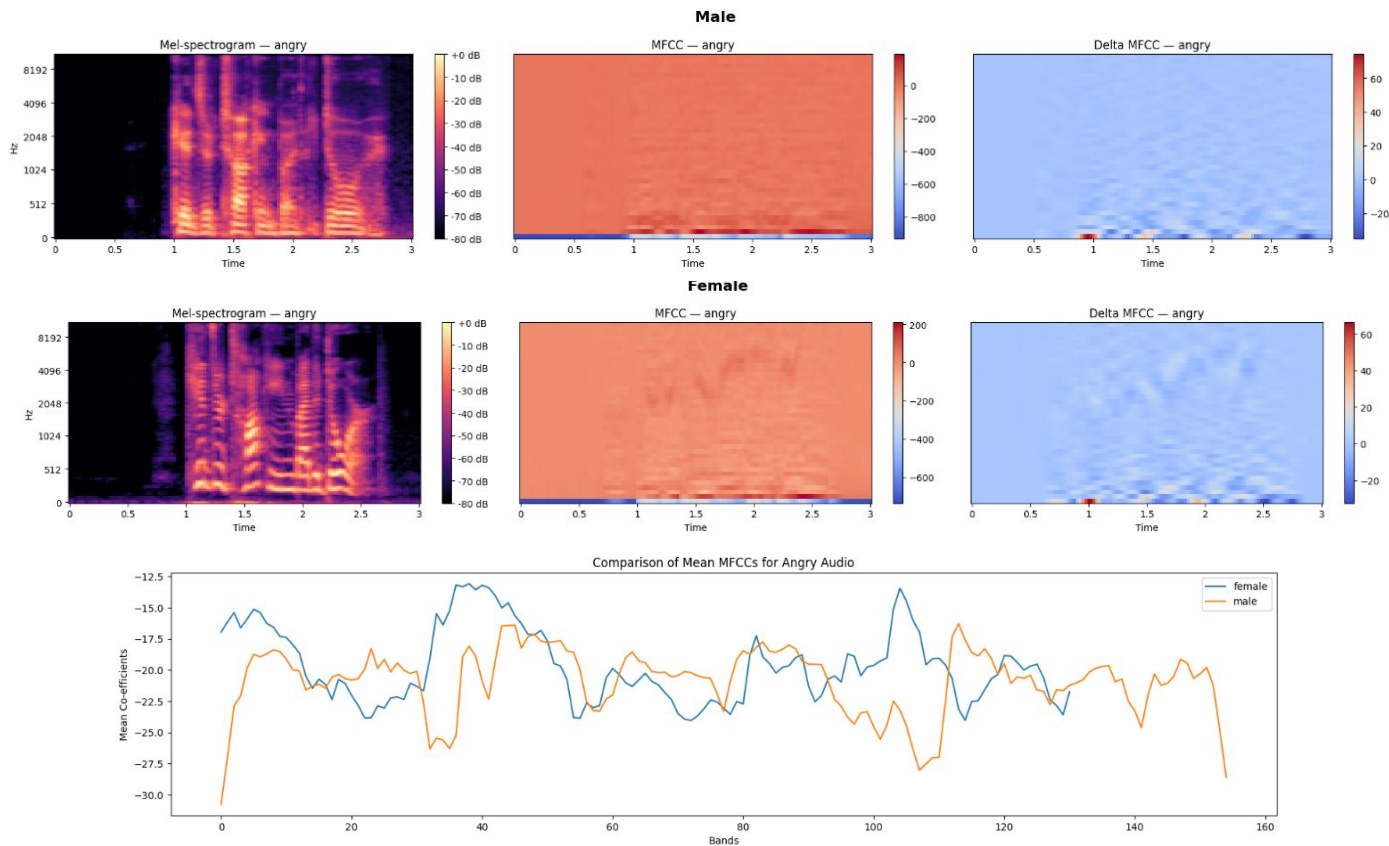


LSTM Pipeline: MFCC Features

Mel spectrogram compression: 128 frequency bands → 40 coefficients | **Shape (40, 130)**

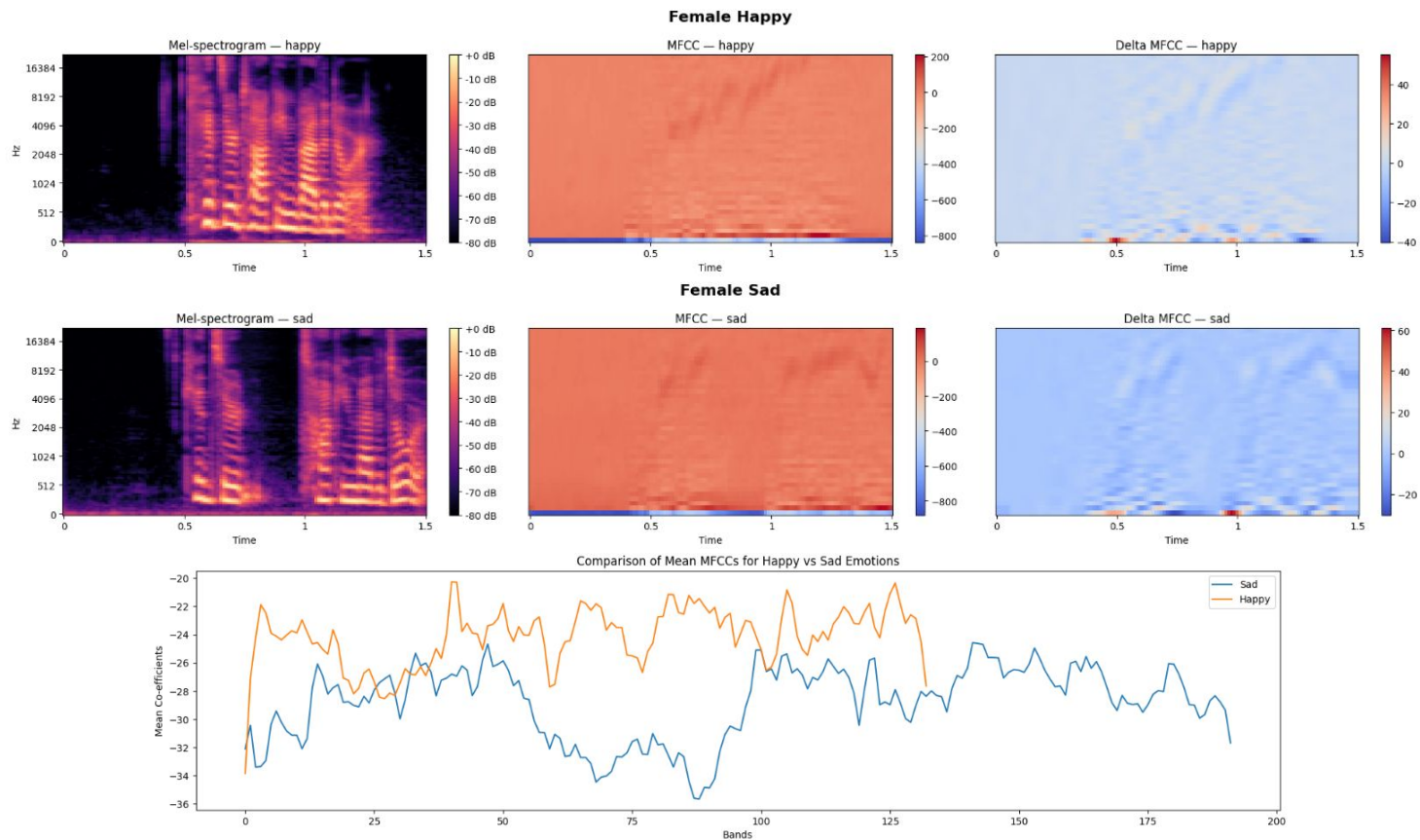
The Data

Comparing Actors



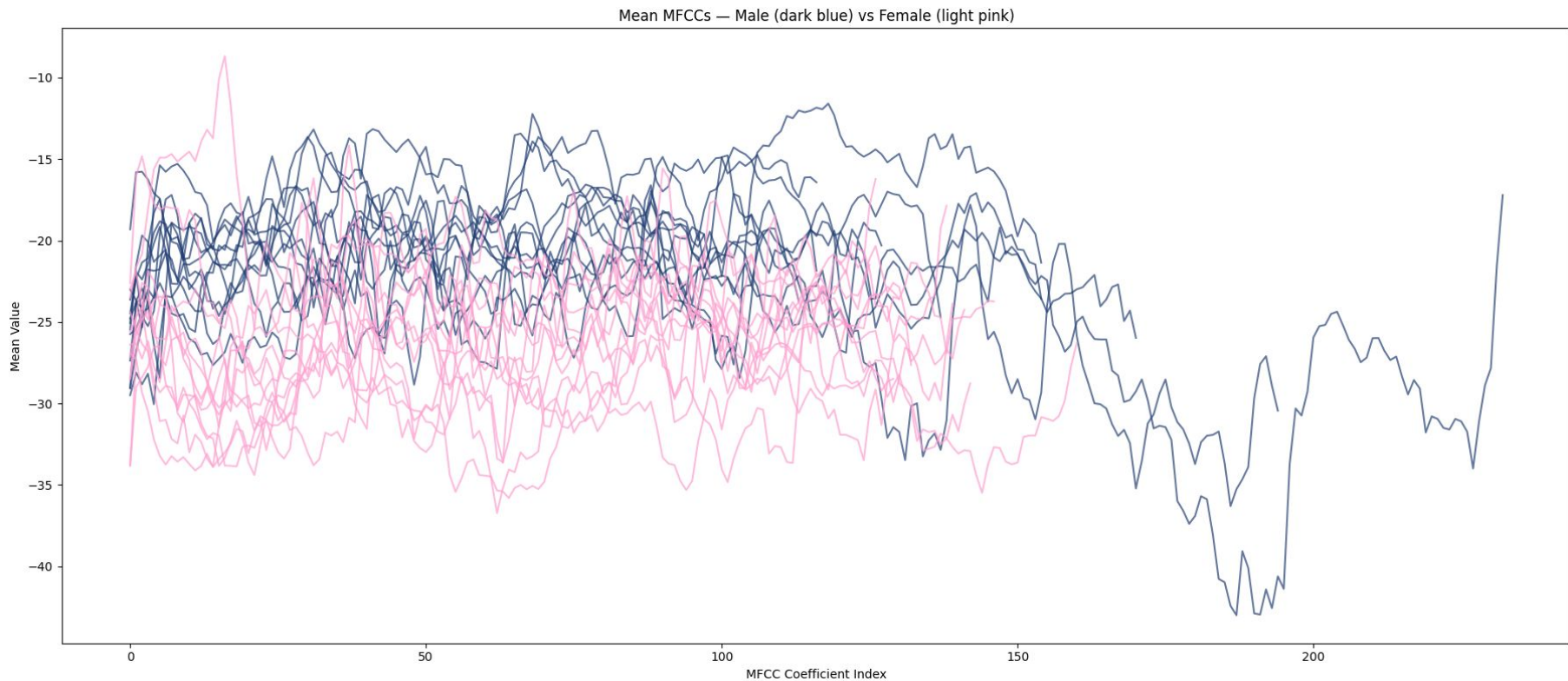
The Data

Comparing Emotions



The Data

Comparing Gender



METHODOLOGY

Baseline — SVM

Methodology

(40, 130) → (40, 1) vectors

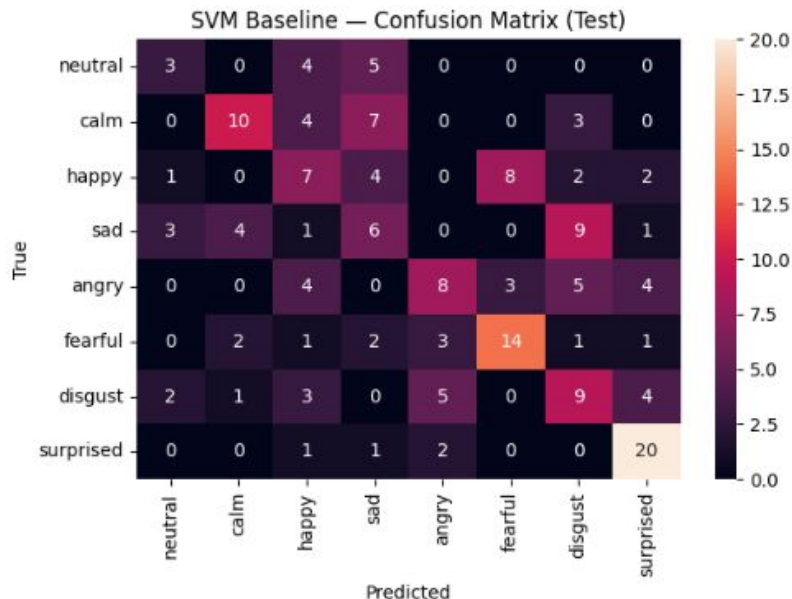
Collapse time into a single value per feature

Results

 **Test accuracy:** 0.428

 **Macro F1:** 0.414

Random guessing accuracy: **~12.5%**



per class F1-score	
neutral	0.29
calm	0.49
happy	0.29
sad	0.24
angry	0.38
fearful	0.57
disgust	0.34
surprised	0.71

CNN Architecture for Emotion Recognition

Architecture Details

- **Input:** (B, 1, 128, 130)
- **Core:** 4-conv layers + Spatial attention mechanism
- **Classifier:** FC layer for emotion mapping
- **Preprocessing:** mel spectrograms

Table 1
FCNN architecture.

Layers
conv2D 32 3 × 3 1
Activation relu
Max Pooling 2 × 2 2
conv2D 64 3 × 3 1
Activation relu
Max Pooling 2 × 2 2
conv2D 16 3 × 3 1
Activation relu
Dropout 0.5
conv2D 7 3 × 3 1
Global Average Pooling
Activation softmax

$$B = \text{sigmoid}[\text{conv2D}_{\text{att}}(A)]$$

$$C = A * B$$

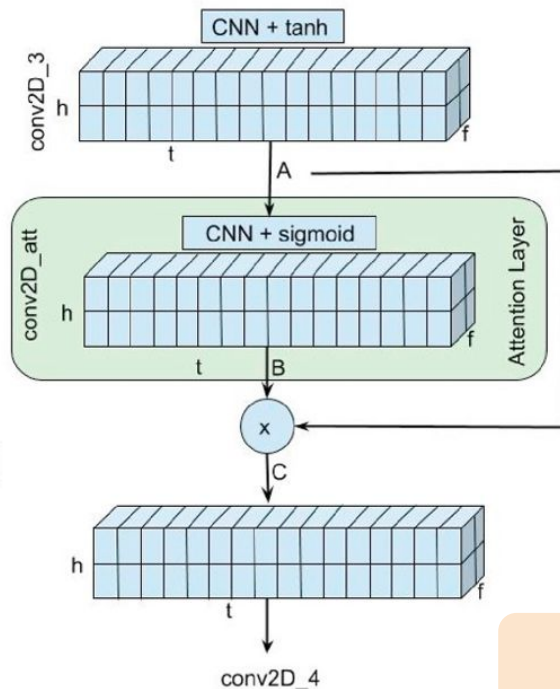


Fig. 6. Attention mechanism.

Architecture:
based on [5]

Baseline:
epochs, learning rate,
optimizer

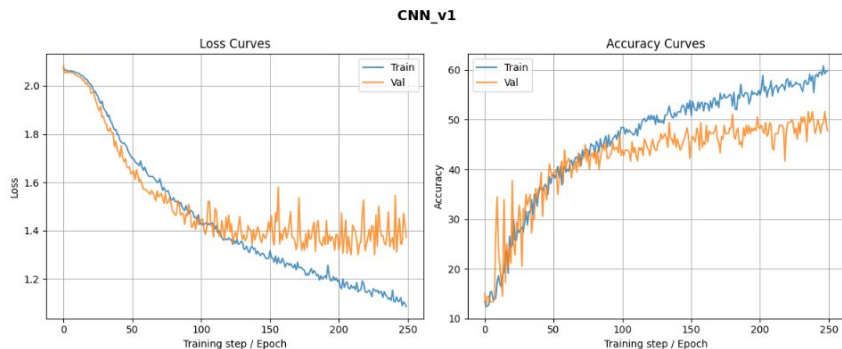
Ablation study:
dropout, attention
Tanh $\rightarrow$ ReLU

Additional:
BatchNorm,
normalization,
augmentation

Differences:
No Augmentation, Mel
Spectrogram

Architecture Choices

Original Architecture



Adam ($\text{lr}=1\text{e-4}$)

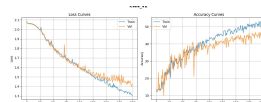
200 epochs

Peak Acc: 51.11%

Discarded for convergence and early stopping:

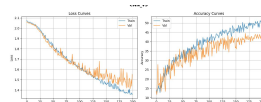
- $\text{lr}=1\text{e-3}, 1\text{e-5}$ (no convergence)
- 250 epochs (train for only 200)
- SGD+Momentum (no convergence within 200 epochs)

Ablation and Additions



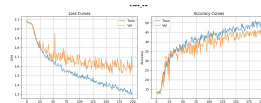
No Dropout

Acc: 46% | Loss: 1.43



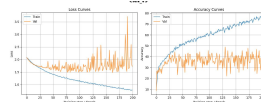
No Attention

Acc: 42% | Loss: 1.44



Tanh $\rightarrow$ ReLU

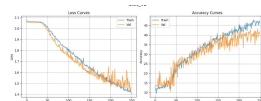
Acc: 45% | Loss: 1.53



BatchNorm

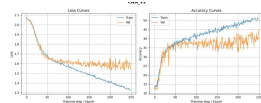
Acc: 48% | Loss: 1.51

overfitting



Normalization

Acc: 44% | Loss: 1.43



Augmentation

Acc: 45% | Loss: 1.52

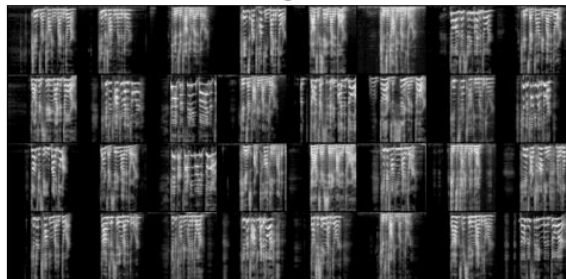
overfitting

Architecture from [5] obtains the best results. They are worse from the paper due to difference in input data and data splits.

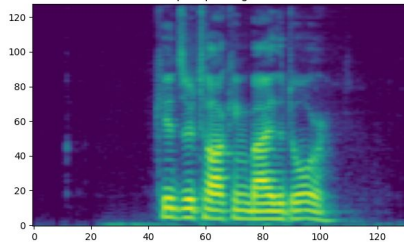
What the CNN sees

Input Data & Attention

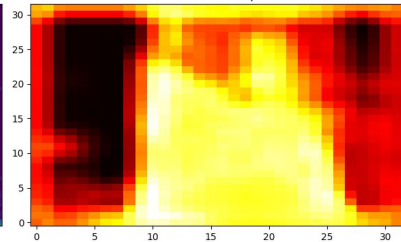
Training Batch



Input spectrogram

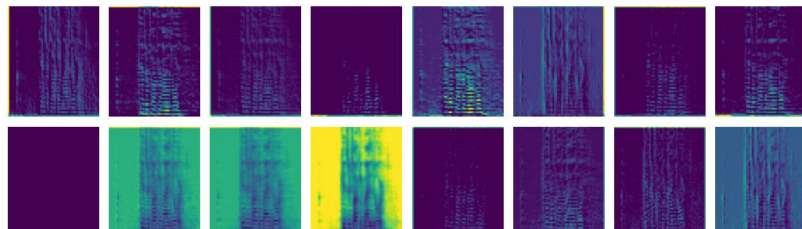


Attention map

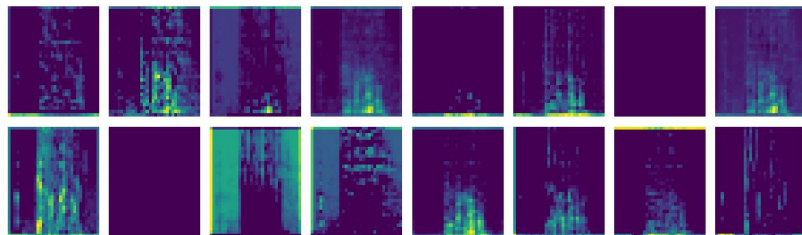


Feature Map Visualizations

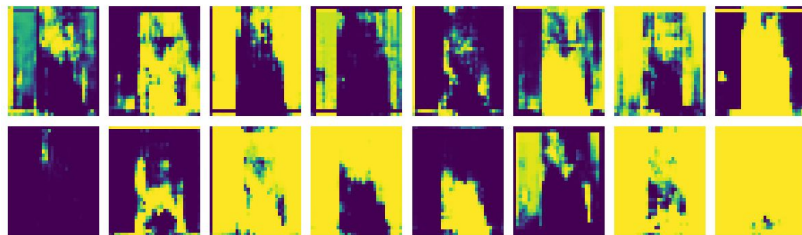
Feature maps after conv1 (first 16 channels)



Feature maps after conv2 (first 16 channels)



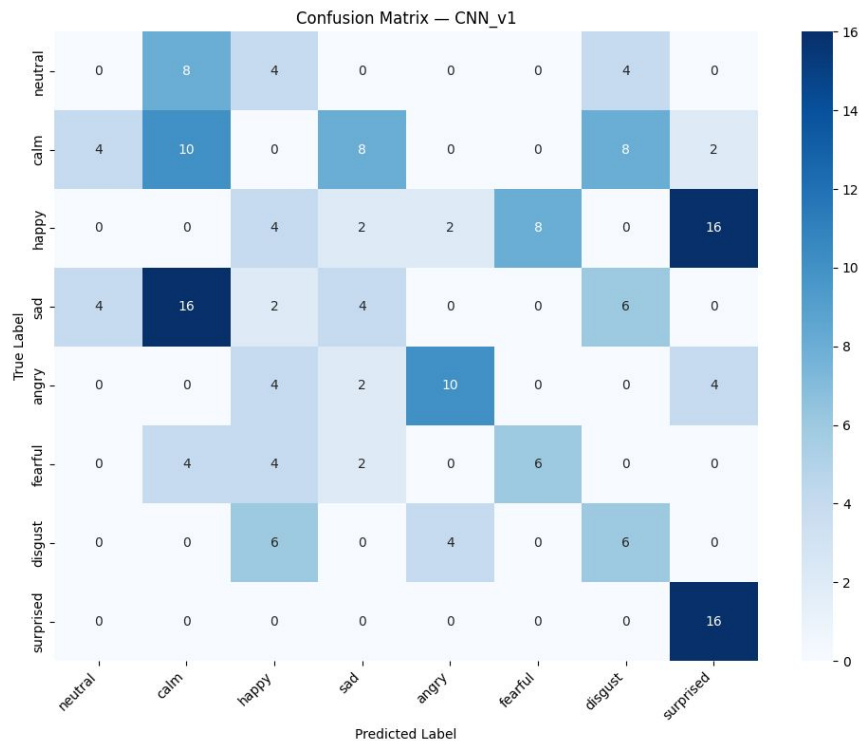
Feature maps after conv3 (first 16 channels)



What the CNN is learning

 **Test accuracy:** 0.311

 **Macro F1:** 0.305



per class F1-score	
neutral	0.00
calm	0.29
happy	0.14
sad	0.16
angry	0.56
fearful	0.40
disgust	0.30
surprised	0.59

LSTM Architecture for Emotion Recognition

Architecture Details

- **Input:** MFCC sequence (130 steps × 40 coefficients)
- **Core:** 2-layer LSTM → last hidden state
- **Classifier:** Linear layer for emotion mapping
- **Preprocessing:** Features normalized per sample

Emotion unfolds over time, by using LSTM, the model reads the sequence step by step and summarises it into a single hidden state for classification.

Model 1 - Standard LSTM

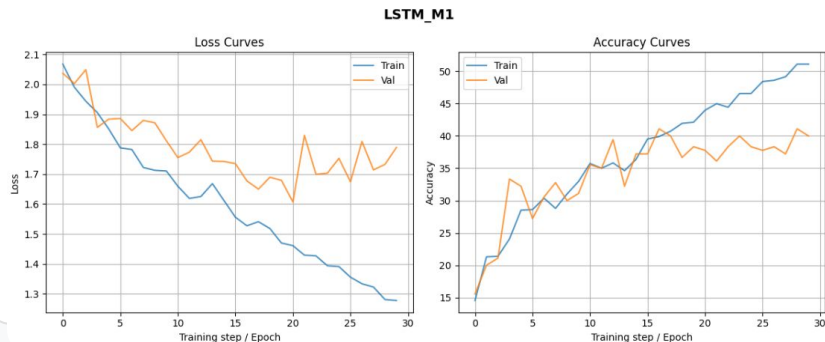
- **Input size:** 40
- **Hidden size:** 128
- **Loss Function:** CrossEntropyLoss()
- **Optimizer:** Adam with learning rate = $1e-3$ and weight decay $1e-4$

Model 2 - Bidirectional + Dropout

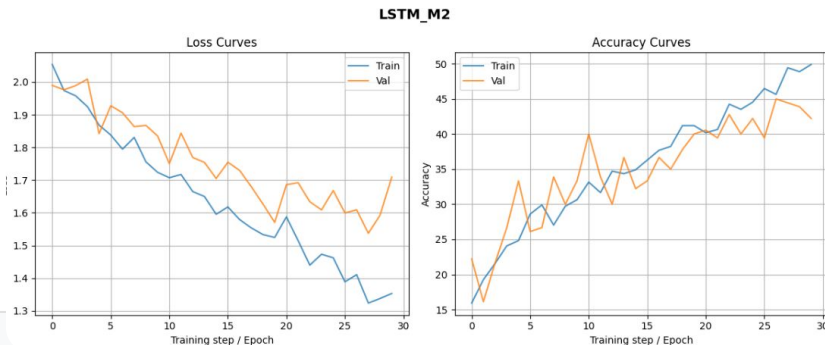
- **Input size:** 40
- **Hidden size:** 128
- **Loss Function:** CrossEntropyLoss()
- **Optimizer:** Adam with learning rate = $1e-3$ and weight decay $1e-4$
- **Dropout:** 0.3

LSTM Results

Model 1 (Standard)



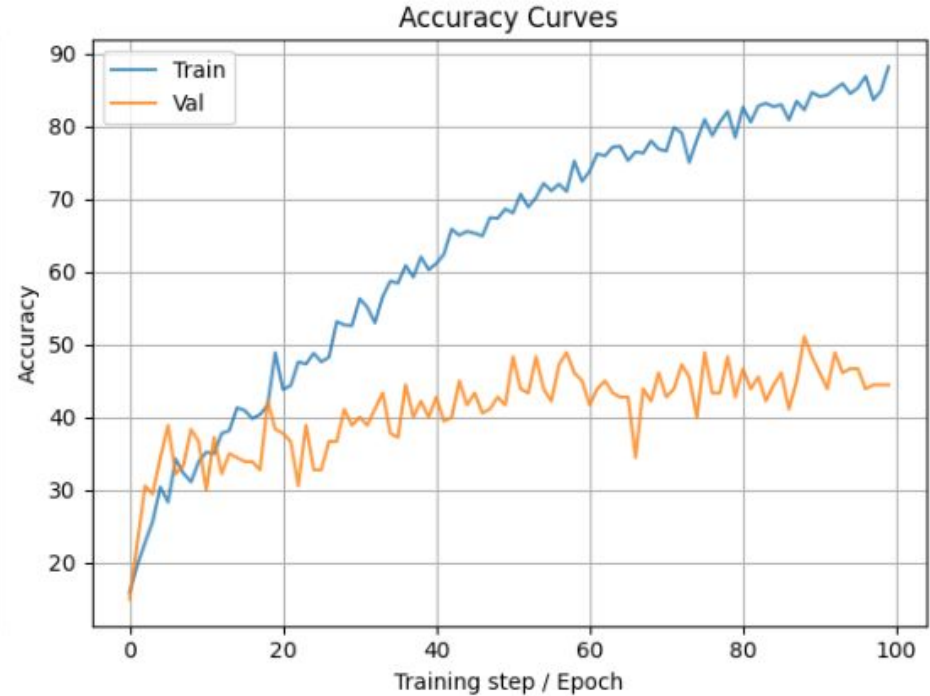
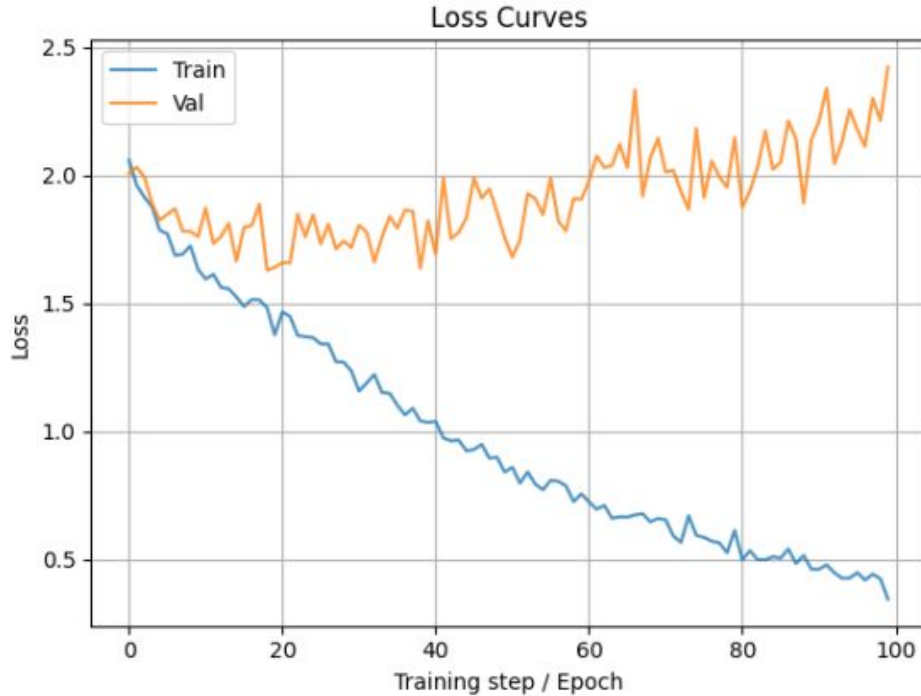
Model 2 (Bidirectional+Dropout LSTM)



Observations

- **Model Comparison:** M1 reaches ~40% validation accuracy, but a gap is already opening. M2 shows a smoother curve and modest improvement to ~44% with a smaller gap.
- **Overfitting Trend:** Across 60/100 epoch runs, validation accuracy flattens(40-45%) while training continues to improve, indicating clear overfitting as capacity increases without better generalization.
- **Data Constraint:** The noisy, oscillating validation loss suggests 180 samples is insufficient for a stable signal, making the optimization landscape difficult to navigate.

LSTM_M1

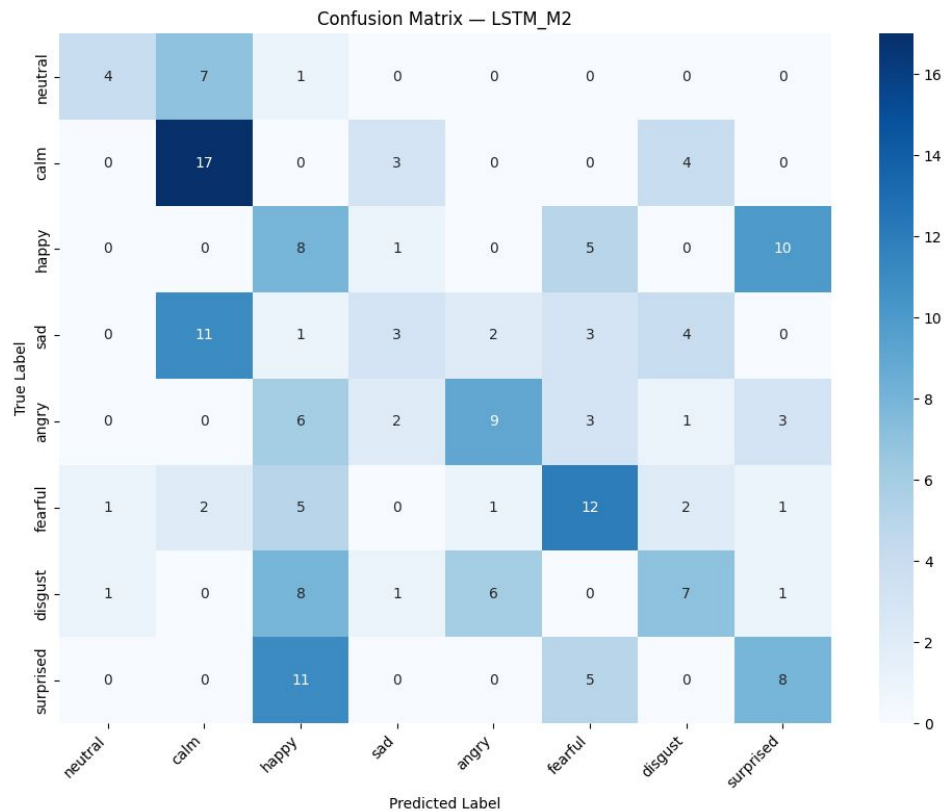


- **Overfitting Trend:** Across 60/100 epoch runs, validation accuracy flattens (40-45%) while training continues to improve, indicating clear overfitting as capacity increases without better generalization.
- **Data Constraint:** The noisy, oscillating validation loss suggests 180 samples is insufficient for a stable signal, making the optimization landscape difficult to navigate.

What the LSTM is learning

 **Test accuracy:** 0.37

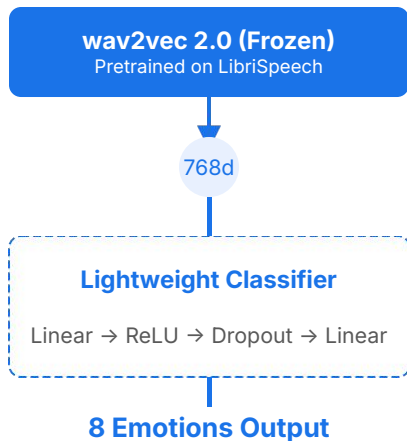
 **Macro F1:** 0.374



per class F1-score	
neutral	0.44
calm	0.56
happy	0.25
sad	0.18
angry	0.43
fearful	0.46
disgust	0.33
surprised	0.34

wav2vec 2.0 — Pretrained Features Fix the Data Problem

Architecture Details



Key Facts

- **960h LibriSpeech:** Pretrained using self-supervised learning (no labels).
- **Frozen Weights:** Feature extractor is locked; only the classifier is optimized.
- **Mean Pooling:** Embeddings compressed over time to (1080, 768) matrix.
- **High Efficiency:** ~0.8 min total training time for extraction and classification.



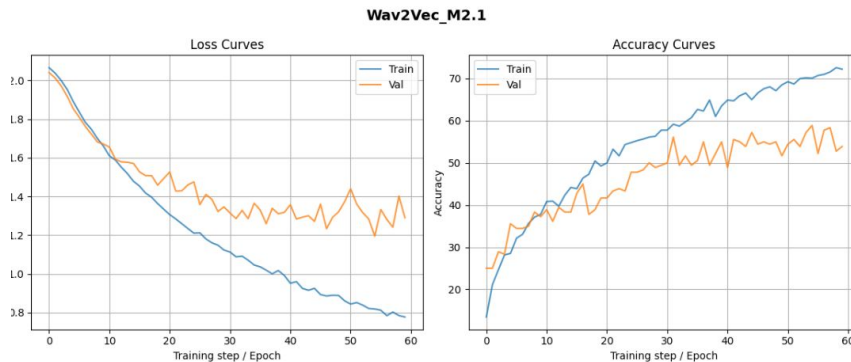
Why this works:

The pretrained model already understands acoustic structure (phonemes, prosody, speaker variation). Our classifier only needs to learn which patterns map to which emotion, a much easier task.

wav2vec 2.0 — Clear Improvement Over LSTM and CNN

Model	Optimiser	Val Acc	Test Acc	F1
Wav2Vec_M1	Adam (lr=1e-2)	56.1%	49.4%	43.6%
Wav2Vec_M2.1	SGD + mom	57.2%	46.7%	43.8%

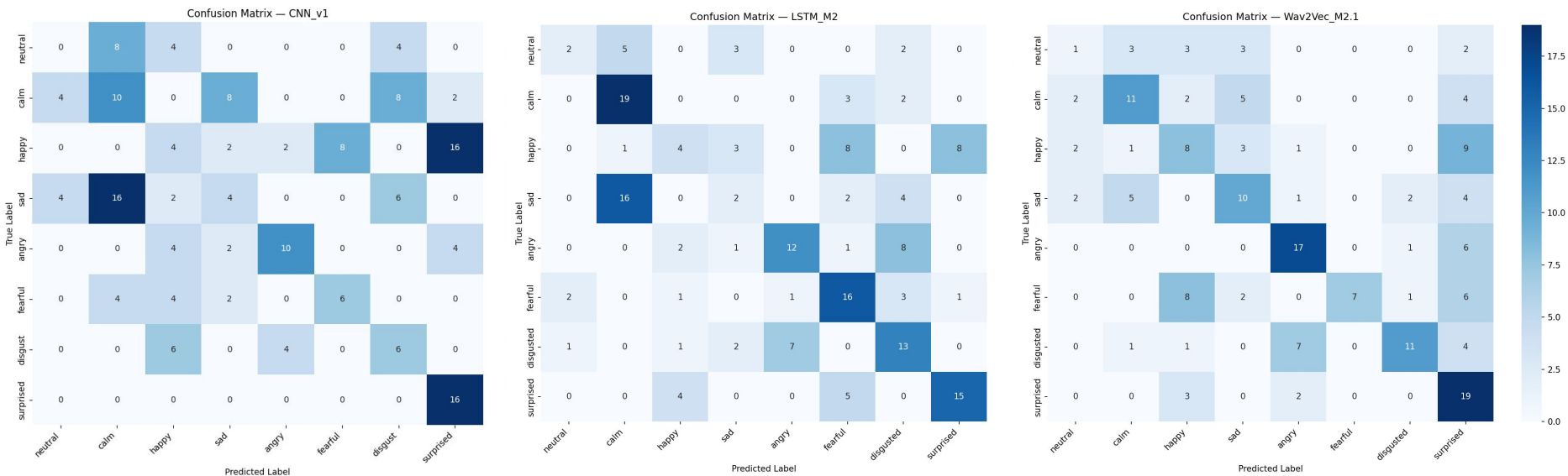
Training Metrics (Best Model)



Observations

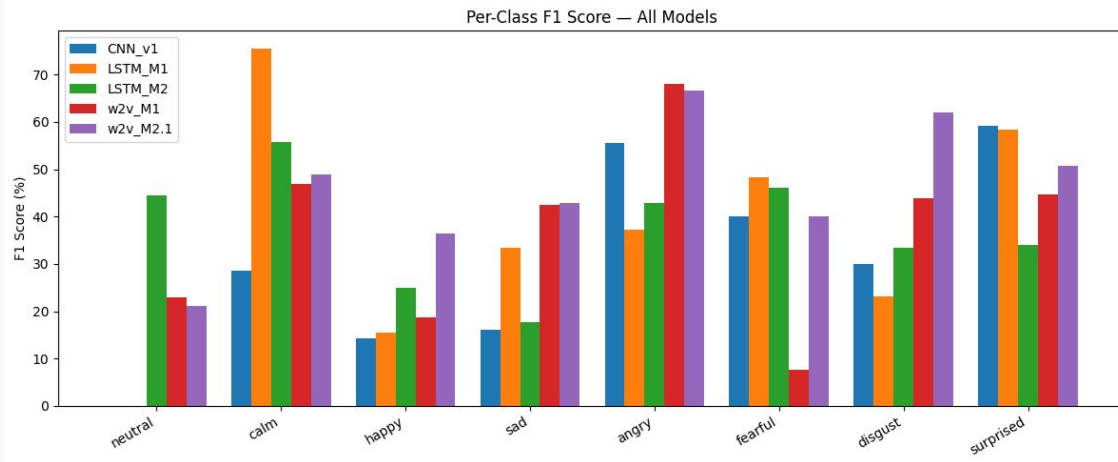
- **Reduced Overfitting:** Smaller train-validation gap than LSTM suggests some generalisation.
- **Validation Peak:** Achieved **57% accuracy**, significantly outperforming LSTM's ~44%.
- **Test Performance:** Consistent ~49–50% results despite the limited RAVDESS dataset size.
- **Gap to State Of The Art:** Still below fine-tuned results (~75%) due to frozen feature extraction approach.

What Errors Are Models Making?



Comparison - Test Results

Model	Accuracy	Macro F1
CNN_V1	31.11%	30.46%
LSTM_M1	42.22%	36.40%
LSTM_M2	37.77%	37.40%
Wav2Vec_M1	40.00%	36.87%
Wav2Vec_M2.1	48.33%	46.05%



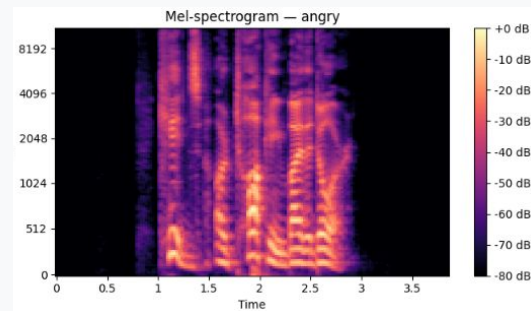
Why did the LSTM beat the CNN?

The **CNN** focuses on **static features**: it tried to learn how each actor sounds (pitch, voice colour).

The **LSTM** focuses on **temporal dynamics**: it learned how emotions change over time, which is universal across people.

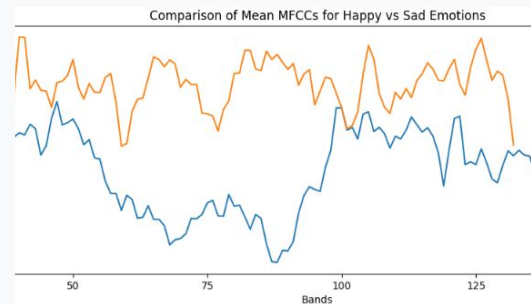
"LSTM learns how emotions change – CNN learns who is speaking."

CNN: Detailed "Image"



Test Accuracy: 31%

LSTM: Time Pattern



Test Accuracy: 41%

CONCLUSIONS



The Data is Key

The amount, preprocessing, and deep understanding are the foundation.



LSTM Superiority

LSTM understands temporal information better than Mel spectrograms and CNNs that in contrast, learn visual patterns more closely related to who is speaking.



Challenge

Speaker-independent splits are difficult to manage, especially with limited data.

References

- [1] Singh, P.; Srivastava, R.; Rana, K.; Kumar, V. A multimodal hierarchical approach to speech emotion recognition from audio and text. *Knowl.-Based Syst.* 2021, 229, 107316. [\[CrossRef\]](#)
- [2] Pepino, L.; Riera, P.; Ferrer, L. Emotion Recognition from Speech Using wav2vec 2.0 Embeddings. *In Proceedings of the Interspeech 2021*, Brno, Czech Republic, 2021; pp. 3400–3404. [\[CrossRef\]](#)
- [3] Mustaqeem; Kwon, S. Att-Net: Enhanced emotion recognition system using lightweight self-attention module. *Appl. Soft Comput.* 2021, 102, 107101. [\[CrossRef\]](#)
- [4] Atila, O.; Sengür, A. Attention guided 3D CNN-LSTM model for accurate speech based emotion recognition. *Appl. Acoust.* 2021, 182, 108260. [\[CrossRef\]](#)
- [5] Wijayasingha, L.; Stankovic, J.A. Robustness to noise for speech emotion classification using CNNs and attention mechanisms. *Smart Health* 2021, 19, 100165. [\[CrossRef\]](#)